JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 90, NO. B14, PAGES 12,523–12,535, DECEMBER 10, 1985

Tectonic Implications of Paleomagnetic Poles From Lower Tertiary Volcanic Rocks, South Central Alaska

JOHN W. HILLHOUSE, C. SHERMAN GROMMÉ, AND BELA CSEJTEY, JR.

U.S. Geological Survey, Menlo Park, California

We have determined the paleolatitude of lower Tertiary volcanic rocks in southern Alaska to measure possible poleward translation of the Wrangellia and the Peninsular terranes after 50 m.y. ago. Previous paleomagnetic studies have shown that in Triassic and Jurassic time these terranes were located near the equator and have moved at least 3000 km poleward relative to the North American craton. Our sample localities are in the northern Talkeetna Mountains in mildly deformed andesite and dacite flows (50.4, 51.3, 53.9, and 56.3 m.y. by K-Ar) that overlap Lower Cretaceous flysch, Lower Permian volcanic rocks of Wrangellia, and Upper Triassic pillow basalt of the Susitna terrane. Results from 26 cooling units (23 of reversed polarity and 3 of normal polarity) give a mean paleomagnetic pole at 69.5°N, 179.6°E, $\alpha_{95} = 12.2°$. Stratigraphic sections from opposite limbs of a syncline yield directional paths that pass the fold test, satisfying a necessary condition for primary origin of the magnetization. The corresponding mean paleolatitude (76°N) of the northern Talkeetna Mountains is $8° \pm 10°$ higher than the latitude predicted from the Eocene reference pole for North America. Therefore, northward drift of the Talkeetna superterrane, which is the amalgamation of the Wrangellia and Peninsular terranes during and after Middle Jurassic time, was probably complete by 50 m.y. ago. Our results are consistent with paleomagnetic poles from uppermost Cretaceous and Paleocene volcanic sequences in Denali National Park, the Lake Clark region, northern Bristol Bay region, and near McGrath. These poles generally lie south of the cratonic poles, suggesting that the region between the Kaltag, Bruin Bay, and Castle Mountain faults has rotated counterclockwise relative to North America since the early Eocene.

INTRODUCTION

Plutons and volcanic rocks of early Tertiary age are abundant in the Alaska Range and Talkeetna Mountains of south central Alaska (Figure 1). These igneous rocks overlap or intrude major accretionary terranes that make up much of northwestern North America [*Jones et al.*, 1981]. Paleontologic and paleomagnetic evidence from three such terranes, Wrangellia [*Jones et al.*, 1977; *Hillhouse*, 1977], Chulitna [*Jones et al.*, 1980], and Peninsular [*Stone and Packer*, 1979], demonstrate that the terranes accreted in Alaska after being transported from original Mesozoic positions far to the south. In studying the lower Tertiary volcanic rocks of southern Alaska, our main objective was to determine whether these terranes have undergone substantial drift relative to North America during the Tertiary.

Paleomagnetic measurements from Paleocene volcanic rocks of the region were first reported by *Helsley* [1970], who sampled the Cantwell Formation at two localities within Denali National Park. Although the age of the Cantwell is now considered to be Paleocene on the basis of isotopic ages and plant fossils, the formation's age was thought to be Early Cretaceous at the time of Helsley's study. The results seemed to indicate that Alaska has rotated at least 20° counterclockwise relative to North America, in apparent agreement with *Carey*'s [1955] orocline hypothesis for the arcuate form of southern Alaska. Subsequent studies of Triassic and Jurassic rocks of the Alaska Peninsula and the eastern Alaska Range [*Packer and Stone*, 1974; *Stone and Packer*, 1977; *Hillhouse*, 1977] revealed unexpectedly low paleolatitudes, and prompted tectonic models calling for exotic origins and large-scale displacements of Alaskan terranes. The Jurassic results of Packer and Stone implied clockwise rotation of southwestern Alaska, in the opposite sense to Helsley's preliminary result. However, the discovery of large anomalies in paleolatitude complicated the interpretation of anomalies in magnetic declination and the determination of rotations. First, the ambiguity of assigning geomagnetic polarities to the observed polarities of the Mesozoic magnetic directions introduced uncertainty in determining the sense of rotations. Second, terranes that originated far from Alaska may have undergone complex rotational histories prior to accretion.

Recent arguments concerning the timing of accretion focus on the geology and paleomagnetism of the part of Alaska that is shown in Figure 2. As indicated by a recent study of the volcanic rocks of the Paleocene Cantwell Formation [*Hillhouse and Grommé*, 1982], terranes north of the Denali fault were essentially in place relative to North America by 60 m.y. ago, at least within the confidence limits of the paleolatitudes. In contrast, results from early Tertiary sites on the Alaska Peninsula and in the southern Talkeetna Mountains imply that terranes immediately south of the Denali fault had moved some 30° (3300 km) poleward in the Tertiary [*Stone et al.*, 1982]. In particular, one site in Paleocene and Eocene sedimentary rocks of the Matanuska Valley yielded a very low paleolatitude, implying that a Tertiary suture having thousands of kilometers of closure lies concealed between Matanuska Valley and the outcrop area of the Cantwell Formation. In order to test this hypothesis, we have extended the sampling of lower Tertiary volcanic rocks on a 75-km transect from the Talkeetna River northward to Denali National Park. The transect crosses Wrangellia, a Jurassic and Cretaceous flysch basin, and several smaller terranes within the highly deformed flysch. Preliminary results of this study were reported by *Hillhouse and Grommé* [1983].

GEOLOGY

Pre-Tertiary rocks of the region exhibit extreme diversity in age, geologic environment, metamorphic grade, and deformational fabric. Assigning these diverse assemblages to terranes (Figure 1), each having a distinguishable stratigraphy, is helpful in understanding the geologic history of southern Alaska. Three terranes of major areal extent within the region

This paper is not subject to U.S. copyright. Published in 1985 by the American Geophysical Union.

Paper number 5B5531.

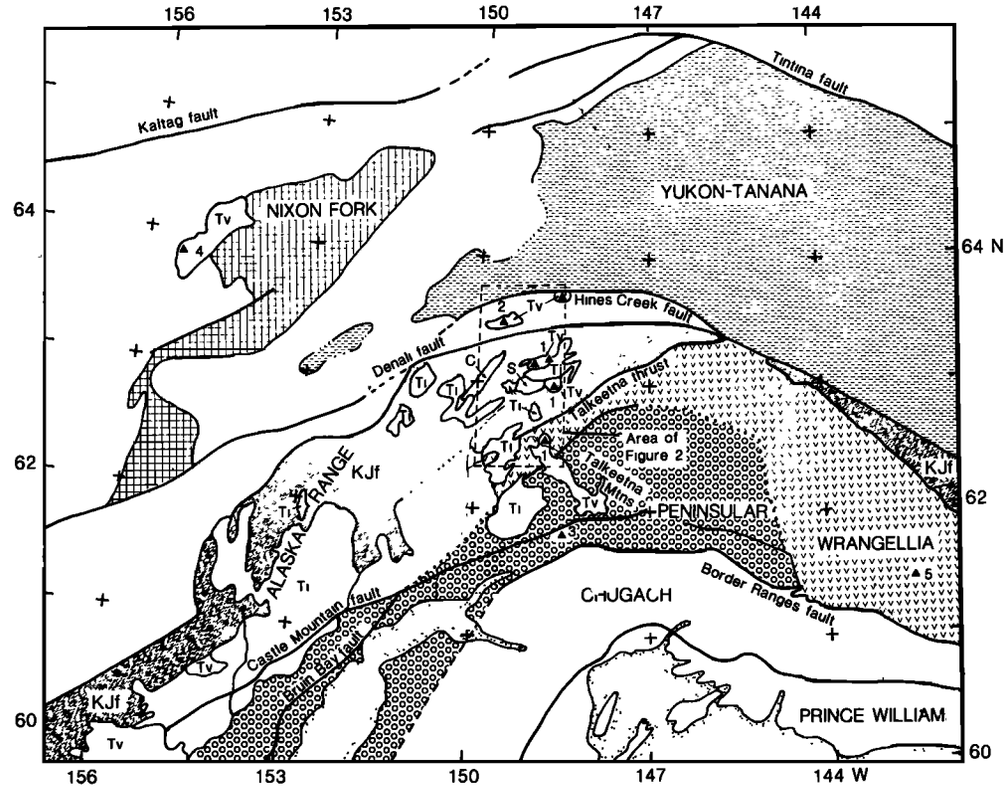


Fig. 1. Map of selected terranes of south-central Alaska. (*C*) Chulitna terrane, (*S*) Susitna terrane, (*KJf*) Upper Jurassic and Cretaceous flysch belt. Early Tertiary intrusive rocks (*Ti*) and volcanic rocks (*Tv*) cross terrane boundaries. Triangles indicate paleomagnetic investigations: (*1*) this study, (*2*) Paleocene Cantwell Formation [*Hillhouse and Grommé*, 1982], (*3*) Chickaloon Formation [*Stone*, 1984], (*4*) Volcanic rocks of the Paleocene Nowitna unit [*Plumley*, 1984], (*5*) Triassic Nikolai Greenstone [*Hillhouse*, 1977].

were described by *Jones et al.* [1981, 1982]. The Yukon-Tanana terrane, most of which lies north of the Hines Creek fault, is primarily composed of muscovite-quartz schist and gneissic granitic rocks. The two major accretionary terranes south of the Denali fault are Wrangellia, dominated by late Paleozoic limestone and Triassic basalt, and the Peninsular terrane, primarily composed of Jurassic batholiths and volcaniclastic rocks. Because Middle Jurassic sedimentary rocks overlap the basements of the Wrangellia and Peninsular terranes, the two terranes were combined under the name "Talkeetna superterrane" to reflect their common translational history after the Early Jurassic [*Csejtey et al.*, 1980, 1982]. Wrangellia is bounded on the northwest by a steeply to moderately dipping fault zone, the Talkeetna thrust [*Csejtey et al.*, 1978, 1982]. *Jones et al.* [1982] described a number of smaller terranes that lie between the Talkeetna thrust and the Hines Creek fault. Of this group the Chulitna terrane has been studied in most detail [*Jones et al.*, 1980]. Most of the smaller terranes appear to be rootless nappes caught up in highly deformed flysch of Late Jurassic and Early Cretaceous age.

Interpretations of the regional geology favor the flysch as the zone of convergence between the northward moving Talkeetna superterrane and the Yukon-Tanana terrane [*Csejtey et al.*, 1978, 1980, 1982], although the original position of the now-collapsed flysch basin has not been established paleomagnetically. Within the Jurassic and Cretaceous flysch, intense deformation and the presence of fragmented terranes attest to the importance of the flysch in the accretionary history. The youngest flysch beds are Cenomanian, indicating that the accretion and accompanying deformation were completed during or after early Late Cretaceous time. Paleolatitude determinations indicate that in the Late Triassic the Wrangellia terrane was at least 28° lower in latitude than it is today relative to North America [*Hillhouse and Grommé*, 1984; *Stone*, 1981]. Jurassic sedimentary rocks of the Peninsular and Wrangellia terranes also yield unusally low paleolatitudes [*Stone et al.*, 1982]. Little is known about the paleopositions of terranes north of the Denali fault in pre-Cretaceous time, although preliminary paleomagnetic results from Ordovician limestone of the Nixon Fork terrane indicate little, if any, movement relative to the craton [*Plumley*, 1984; *Plumley and Coe*, 1983]. Therefore, the region generally north of the Denali fault is evidently the backstop for the northward drifted terranes.

The Late Cretaceous and Paleocene were times marked by widespread igneous activity in the central Alaska Range and Talkeetna Mountains, and by orogenic activity in the narrow zone roughly bounded by the Denali and Hines Creek faults (Figure 2). In this zone, coarse sediments and volcanic rocks of the Paleocene Cantwell Formation were deposited in a continental basin [*Wolfe and Wahrhaftig*, 1970; *Hickman and Craddock*, 1976]. The upper part of the Cantwell Formation consists of calc-alkaline volcanic flows and pyroclastic rocks that were renamed the Teklanika Formation by *Gilbert et al.* [1976]. Potassium-argon age determinations from the volcanic rocks range from 41.8 to 60.6 m.y., in general agreement with Paleocene plant fossils reported by *Wolfe and Wahrhaftig* [1970] from the Cantwell Formation. These dated volcanic rocks were the target of a paleomagnetic study [*Hillhouse and Grommé*, 1982], which found no statistically significant differ-

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ence between the paleomagnetically determined latitude and the latitude calculated for the region from cratonic reference poles. During or soon after deposition, the Cantwell Formation was folded along east axes and broken by numerous normal faults. The age of major deformation is constrained by an overlapping hornblende andesite flow, dated at 43.2 m.y. by the potassium-argon method [*Gilbert*, 1976]. Folded Cantwell beds are also overlapped by less-deformed Oligocene and Miocene coal-bearing beds in Denali National Park [*Wahrhaftig*, 1958; *Wahrhaftig et al.*, 1969; *Hickman and Craddock*, 1976].

South of the Denali fault, Late Cretaceous and early Tertiary igneous rocks are less deformed than the Cantwell Formation. Granodiorite and tonalite plutons intrude the folds and the faulted boundaries of all of the Mesozoic terranes. For example, the Ruth pluton, which yielded biotite K-Ar ages of 55.7, 56.2, and 56.5 m.y. [*Reed and Lanphere*, 1972; *Reed and Nelson*, 1980], cuts the Jurassic and Cretacous flysch and the southern margin of the Chulitna terrane. Other plutons that range in age from 56.3 to 58.6 m.y. cut the Susitna terrane, the Talkeetna thrust, and the fault that separates the Peninsular terrane from Wrangellia [*Csejtey et al.*, 1978]. The plutons are associated with andesite flows, pyroclastic rocks, and dacite bodies of Early Eocene or possibly Paleocene age, according to several K-Ar determinations: 50.4 ± 2.0 m.y. (hornblende), 51.3 ± 2.5 m.y. (whole rock), 56.3 ± 2.5 (whole rock), and 53.9 ± 1.6 m.y. (sanidine). Analytical data for the whole rock and hornblende age determinations are tabulated by *Csejtey et al.* [1978]. These extrusive rocks were the target of our paleomagnetic study (Figure 2). The sample localities are in flows that overlap and rest unconformably on the Jura-Cretaceous flysch, Triassic pillow basalts of the Susitna terrane, and Lower Permian rocks of Wrangellia. Deformation of the Eocene volcanic rocks is limited to broad, open folds having dips no greater than 25°. Mid-Tertiary sedimentary rocks are sparse and poorly exposed in this region, so narrow constraints cannot be placed on the timing of deformation.

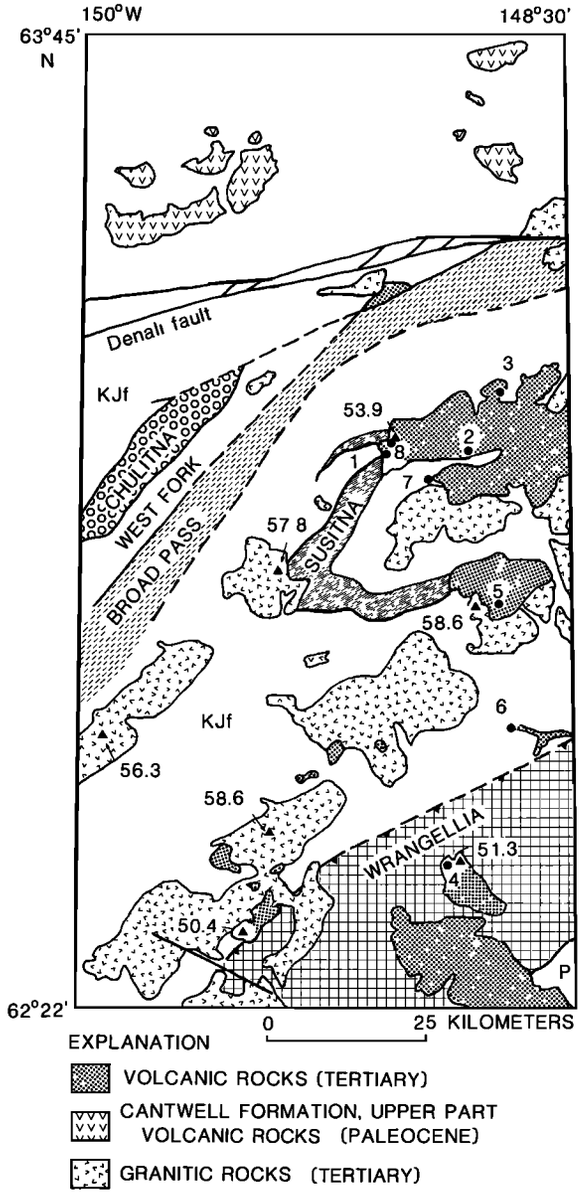


Fig. 2. Outline geologic map with selected terranes showing localities dated by potassium-argon (triangles) and paleomagnetic sample sites (dots) keyed to Tables 1 and 2. Ages given in m.y. (*KJf*) Upper Jurassic and Cretaceous flysch, (*P*) Peninsular terrane.

TABLE 1. Paleomagnetic Sample Sites in the Talkeetna Mountains, Alaska

Site Number	Number of Units	Latitude, °N	Longitude, °E	Rock Type	Bedding	
					Strike	Dip
1	5	63.16	210.94	Andesite	298.0	16.0 N
2	1	63.17	211.19	Dacite	302.0	15.0 NE
3	5	63.24	211.29	Andesite	100.0	10.5 S
4	4	62.58	211.11	Andesite	25.0	4.0 E
5	1	62.96	211.25	Dacite	325.0	8.0 SW
6	1	62.78	211.31	Felsite	0.0	0.0
7	2	63.12	211.06	Andesite	307.0	6.0 SW
8	8	63.18	210.94	Andesite	62.0	22.0 S

With the exception of the Denali fault, terrane boundaries in Figure 2 are crosscut by the early Tertiary plutons and partially covered by volcanic rocks that show no apparent offset along the boundaries. In addition, the igneous suite postdates the intense isoclinal folding that is seen in the Upper Jurassic and Lower Cretaceous flysch. Therefore our paleomagnetic results constrain the early Tertiary movements of essentially a single crustal block consisting of the deformed flysch basin, western Wrangellia, and the eastern part of the Peninsular terrane. Post-Eocene faults have cut this block, but the lateral displacements are no more than a few kilometers [*Reed and Nelson*, 1980; *Csejtey et al.*, 1978, 1982]. The emplacement of the smaller terranes, such as Chulitna, into the flysch occurred sometime during middle Cretaceous time.

PALEOMAGNETIC SAMPLING AND LABORATORY TECHNIQUES

Using portable drilling equipment, we collected oriented cores at eight localities in the Eocene volcanic rocks of the northern Talkeetna Mountains (Figure 2, Table 1). We took eight or nine independently oriented samples from each of 27 cooling units, which consisted of andesite flows and one dacite body. The andesite flows, 2 to 20 m thick, occur with thin tuffaceous beds in well-stratified piles. Many of the flows have reddened scoriaceous tops indicating that the eruptions occurred in a subaerial environment. The dacite body, which is several hundred meters thick, crops out in extensive rows of columns separated by cooling fractures perpendicular to the local structure. Sites 8 and 4 are from localities that were dated directly by K-Ar.

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Paleomagnetic Results From Lower Eocene Volcanic Rocks of the Northern Talkeetna Mountains, Alaska

Site (Unit)	N/N_c	Treatment	R	k	α_{95}	I	D	I_c	D_c	λ	ϕ
1(1)	8(8)	40 mT	7.9860	499	2.5	−68.5	139.5	−59.0	166.5	65.3	56.3
1(2)	8(8)	40 mT	7.9828	408	2.8	−52.0	67.5	−62.7	86.5	36.9	147.1
1(3)	8(8)	30 mT	7.9767	301	3.2	−73.4	202.4	−57.4	205.0	60.7	348.0
1(4)	8(8)	40 mT	7.9770	305	3.2	−77.0	134.8	−66.7	175.0	75.9	44.5
1(5)	8(8)	40 mT	7.9765	298	3.2	−77.2	329.6	−56.9	258.1	56.9	265.0
2(1)	8(8)	430°C	7.9861	503	2.5	−72.1	17.9	−85.1	331.3	54.3	219.3
3(1)	4(8)	20–50 mT	3.9758	124	8.3	−79.1	130.2	−79.4	72.4	52.0	178.4
3(2)	2(8)	60 mT	1.9971	340	13.6	−81.1	122.7	−79.2	59.5	48.9	183.5
3(3)	7(8)	20 mT	6.8949	57	8.1	−81.8	89.3	−75.6	44.0	40.3	186.7
3(4)	8(8)	20 mT	7.9602	176	4.2	−85.5	94.7	−78.2	32.3	42.7	195.1
3(5)	8(8)	30 mT	7.9296	99	5.6	−81.9	178.4	−86.9	41.9	58.5	203.5
4(1)	0(8)										
4(2)	7(8)	40–50 mT	6.9896	576	2.5	69.1	81.2	65.6	86.2	42.7	276.8
4(3)	8(8)	40 mT	7.9804	357	2.9	72.9	340.1	75.4	351.0	85.8	124.6
4(4)	8(8)	30 mT	7.9922	899	1.9	74.2	349.4	76.2	2.7	88.3	255.1
5(1)	8(8)	40 mT	7.9830	411	2.7	−77.0	156.5	−73.4	184.3	85.8	0.3
6(1)	6(8)	40 mT	5.7997	25	13.7	−59.9	72.4	−59.9	72.4	28.4	156.2
7(1)	8(8)	30 mT	7.9742	271	3.4	−72.7	32.3	−78.7	29.8	43.0	196.4
7(2)	8(8)	40 mT	7.9670	212	3.8	−68.8	53.5	−74.4	59.5	41.9	176.7
8(1)	8(8)	30 mT	7.9898	684	2.1	−38.4	154.5	−60.4	156.1	64.0	75.1
8(2)	8(8)	30 mT	7.8886	63	7.0	−49.6	175.6	−68.2	196.4	75.3	346.9
8(3)	8(8)	40 mT	7.8067	36	9.3	−44.5	134.9	−64.9	122.4	54.8	121.1
8(4)	8(8)	30 mT	7.9920	872	1.9	−42.6	90.0	−49.3	67.9	17.5	153.0
8(5)	8(8)	40 mT	7.9887	620	2.2	−72.9	157.1	−84.8	315.2	55.2	223.7
8(6)	9(9)	60 mT	8.9445	144	4.3	−56.8	150.6	−78.8	148.1	76.1	156.7
8(7)	8(8)	40 mT	7.9711	242	3.6	−71.1	159.8	−85.9	294.4	58.9	225.6
8(8)	9(9)	40 mT	8.9395	132	4.5	−44.2	134.4	−64.5	121.8	54.0	121.1

Mean paleomagnetic pole of 26 unit VGP's: 69.5°N, 179.6°E, $A_{95} = 12.2°$, Angular s.d. = 32.8°, $R = 22.0561$, $K = 6$.

N_c, number of specimens collected; N, number of specimens averaged; Treatment refers to temperature of thermal demagnetization in degrees C, or applied alternating field in milliTesla; R, resultant vector; k, Fisher precision parameter; α_{95}, 95% confidence limit for directions in degrees; I and D, inclination and declination without corrections for tilt of the bedding, in degrees; I_c, D_c, inclination and declination of tilt-corrected directions, in degrees; λ and ϕ, latitude and east longitude of virtual geomagnetic pole, in degrees; K, Fisher precision parameter for poles; A_{95}, 95% confidence limit for mean pole.

Only those sites having exceptional exposure and well-defined structure were selected for sampling. During the magnetic orientation procedure for each core the effects of local declination anomalies were countered by backsighting to one or more landmarks and applying corrections. As shown in Table 1, most of the dips of the strata are small, so only minor errors in declination are to be expected from application of simple tilt corrections. Sites 1 and 8 are from opposite limbs of an open syncline having a plunge of no more than 10°.

In the laboratory, alternating field (AF) treatments were performed in a shielded, tumbler-type degausser capable of a maximum peak AF of 100 mT (milliTeslas). Thermal demagnetizations were carried out in air in a magnetically shielded furnace in which the ambient direct field was reduced to 2 nT (nanoTeslas). Curie temperatures were measured with an automatic recording thermobalance employing direct fields of 0.25–0.40 T. During heating in the thermobalance the specimens were kept in an argon atmosphere to inhibit oxidation of the magnetic minerals.

PALEOMAGNETIC RESULTS

For the most part, directions of natural remanent magnetization (NRM) were of reversed polarity, and intensities of NRM ranged from 10 to 10^{-2} A/m. We employed the following AF demagnetization procedure to remove unstable components of magnetization from the specimens. Initially, two specimens from each cooling unit were progressively demagnetized in steps up to 100 mT. Stable end point magnetizations were then determined by visual inspection of orthogonal projections of the magnetic vector components. We considered the stable end point to have been isolated when changes in magnetic direction became negligible as the intensity of magnetization decayed toward zero. Following determination of the AF strength that was necessary to isolate the stable end point of the pilot specimens, the remaining specimens were demagnetized at two to four steps in fields above this strength. Results from the AF step that minimized the angular dispersion of the cooling unit were then selected and averaged (Table 2). A comparison of Fisher precision parameters before and after AF treatment showed that the treatment substantially decreased the angular dispersion of directions in all but two of the cooling units.

A few of the cooling units required additional AF treatments to isolate the stable magnetic direction. For these cases in which the pilot specimens either exhibited erratic directional changes throughout the experiment or failed to reach a stable direction, all specimens from the cooling unit were progressively demagnetized up to 100 mT. After inspection of the orthogonal vector diagrams we rejected results from any specimens that failed to reach a stable end point direction. Units from sites 3, 4, and 6 were treated in this way. Unit 1 from site 4 had widely dispersed directions of NRM and maintained an unsatisfactory degree of dispersion despite AF treatment. The unit's anomalously high intensities (10 A/m) of

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

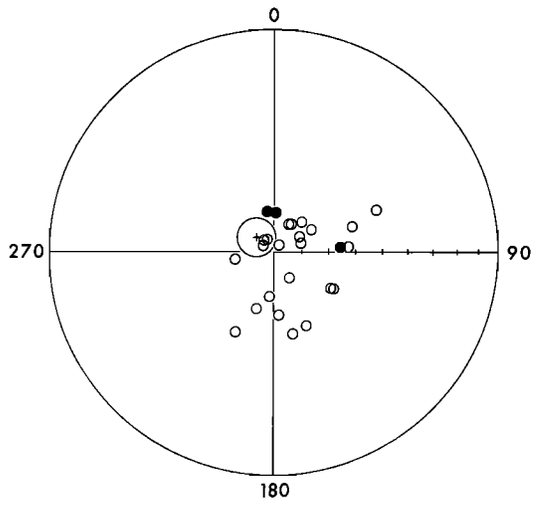


Fig. 3. Magnetic field directions (after AF treatment and corrected for tilt of the bedding) from volcanic units of the northern Talkeetna Mountains. Small circles indicate cooling units of reversed polarity (upper hemisphere); dots indicate normal polarity (lower hemisphere). The mean direction (plus) of 26 cooling units is shown with its α_{95} confidence circle (lower hemisphere). Lambert equal-area projection.

NRM suggested that the outcrop had been struck by lightning, the magnetic effects of which could not be removed. Including results from this lightning-struck unit, the results from 22 specimens were rejected from the total collection of 218.

The columnar dacite of site 2 did not respond well to AF treatment, so we tried thermal demagnetization to improve the grouping of magnetic directions. All specimens from the site were progressively heated up to 500°C. Stable end point magnetizations were isolated by heating the specimens above 300°C, and 430°–433°C was selected as the optimum cleaning step (Table 2). Thermal treatment increased the precision parameter of site 2 from 5 to 500. These extremely hard, glassy rocks were heated during prolonged drilling, so the initially large dispersion of directions may have been an artifact of the coring operation.

After treatment and selection, the data set consists of 23 cooling units with reversed polarity and 3 with normal polarity (Figure 3; Table 2). On average, the angular standard deviation for directions within individual cooling units (s_w) is 5.9°. We consider each unit to be a spot reading of the ancient magnetic field direction. The mean normal polarity direction, obtained by inverting all reversed directions through the origin, is inclination = 82.1°, declination = 310.9°, and α_{95} = 7.0°.

The syncline that was sampled at sites 1 and 8 provides a fold test, a necessary condition for establishing whether the remanence is primary. The north limb of the fold is represented by a stratigraphic sequence of eight andesite flows; the south limb is represented by a sequence of five flows. Application of the tilt corrections significantly decreases the separation of the directional paths of the two sequences (Figure 4). Hence the data satisfy a necessary condition for primary cooling remanence, that the magnetization was acquired before the rocks were folded. The age of folding is only loosely constrained between the Quaternary and the age of the rocks, because mid-Tertiary sedimentary rocks are sparse and poorly exposed in the northern Talkeetna Mountains.

The stack of eight lava flows from site 8 yielded magnetic field vectors that describe a broad loop. Starting from a point in the upper, southeast quadrant, the corrected vectors cross over the vertical pole of the direction diagram, swing 45° out to a point in the upper, northeast quadrant, and then return to the southeast quadrant (Figure 4). This path, which is roughly reproduced at site 1, is a probable record of secular variation of the ancient magnetic field and implies that the stratigraphic section accumulated over hundreds of years. The dual polarity of the whole data set also implies that the collection represents thousands of years of geomagnetic secular variation. Therefore the mean direction from our study is probably a valid representation of the axial, geocentric dipole field in early Eocene time. The mean of all site virtual geomagnetic poles (VGP) is 69.5°N, 179.6°E with an α_{95} of 12.2°.

Compilations of paleomagnetic data from sites distributed worldwide give estimates of the angular dispersion of VGP's to be expected at a particular latitude. The common measure of angular dispersion is the angular standard deviation (S_T)

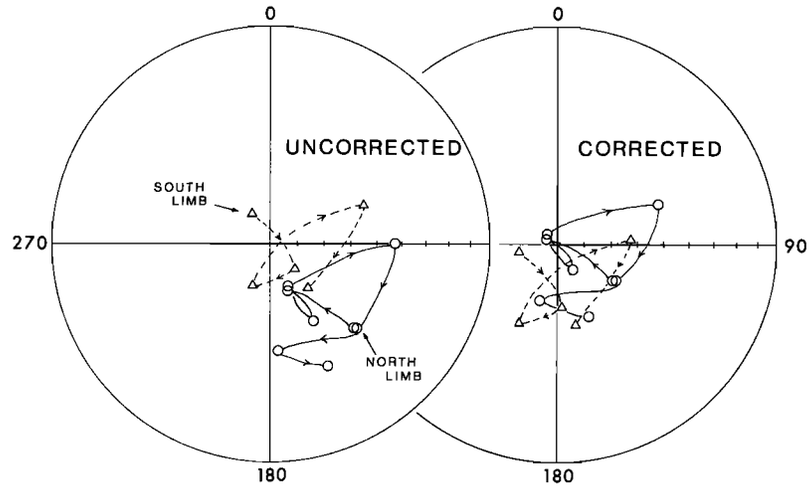


Fig. 4. Comparison of magnetic field directions from the limbs of a syncline in the northern Talkeetna Mountains before (left) and after (right) the directions were corrected for tilt of the bedding. Stratigraphic sequences of lava flows were sampled at site 1 (triangles) and site 8 (circles). All directions have reversed polarity. Lambert equal-area projection, upper hemisphere.

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

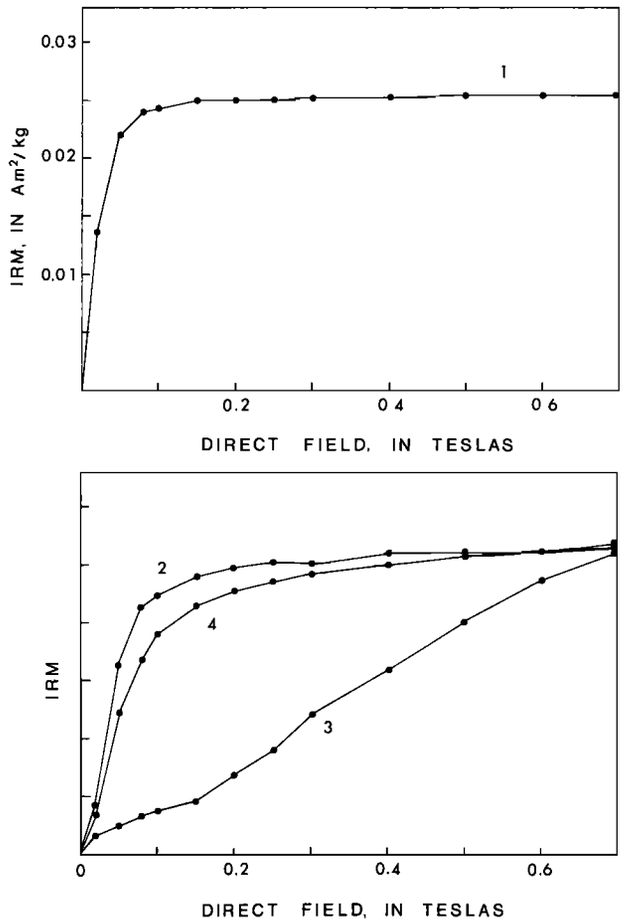


Fig. 5. Isothermal remanent magnetization (IRM) curves for representative specimens. Site designations refer to Table 2 (this study) and Table 1 of *Hillhouse and Grommé* [1982]. (*1*) Site 4(3), Talkeetna Mountains; (*2*) Site 10, Cantwell Formation, 1 division = 10^{-2} Am^2/kg; (*3*) Site 5, Cantwell Formation, 1 division = 10^{-3} Am^2/kg; (*4*) Site 1(5), Talkeetna Mountains, 1 division = 3.3×10^{-3} Am^2;kg.

about the mean VGP. For rocks 5 m.y. old and younger, S_T averages 13.3° at the equator and increases up to 19.8° at latitudes greater than 60° [*McElhinny and Merrill*, 1975]. Wobble of the main dipole and movement of nondipole foci are believed to be the causes of the dispersion. The latitude variation of S_T is mainly due to the geometry of adding dipole and nondipole fields on a sphere and of the transformation of directions to VGP's [*Cox*, 1970]. Our study of early Eocene rocks in the Talkeetnas yielded $S_T = 32.8°$ at a mean paleolatitude of 76°N. This value is unusually high compared to S_T from the high-latitude compilation mentioned above. The cause of this anomaly cannot be due to temporal variations in angular dispersion, because Eocene igneous rocks from Montana (paleolatitude 49°N) give $S_T = 18.8°$, only slightly larger than the value (17.9°) given by *McElhinny and Merrill* [1975] for mid-latitude sites covering the last 5 m.y. [*Diehl et al.*, 1983]. We believe that the unusually high value of S_T in Alaska may be a natural feature of the geomagnetic field because previously reported values from studies in Alaska and Iceland tend to be higher than S_T from the world-wide compilation of *McElhinny and Merrill* [1975], which combines data from the southern hemisphere with these northern studies. For example, paleomagnetic studies of young lavas on Nunivak and the Pribilof Islands of Alaska yielded $S_T = 23.5°$ [*Cox and Gordon*, 1984]. *Harrison*'s [1980] compilation of western Icelandic data yielded $S_T = 24.2°$ when VGP's lying more than 45° from the mean were excluded. If we treat our data in the same manner, by excluding three low-latitude VGP's, we obtain $S_T = 29.1°$. Ninety-five percent limits of this statistic, as calculated by the method of *Cox* [1969], are $24.1° < S_T < 36.7°$. Given the poorly constrained confidence range on our relatively small sample, the difference between S_T of our study and the other high-latitude studies may not be significant. Hence the relatively large angular dispersion is most likely to be a natural feature of the geomagnetic secular variation rather than being due to artificial causes, such as inaccurate dip corrections, unremoved secondary magnetic components, or inaccurate sample orientations.

Rock Magnetism and Origin of Remanence

We determined the rock magnetic properties of representative specimens from the volcanic rocks of the northern Talkeetna Mountains and from the volcanic rocks of the upper part of the Cantwell Formation, the Teklanika Formation of *Gilbert et al.* [1976]. The Teklanika volcanic rocks were included in this study to supplement the paleomagnetic results that were reported by *Hillhouse and Grommé* [1982]. We measured the acquisition of isothermal remanent magnetization (IRM), the median destructive field (MDF) of NRM, Curie temperatures, and unblocking temperatures of NRM. Polished sections from 19 representative specimens were examined with a reflecting light microscope at a magnification of × 1500.

The IRM determinations were made on one specimen from each of the Talkeetna cooling units and from sites 2, 5–11, and 18 in the Teklanika volcanic rocks. The IRM curves, which were carried out to 0.7 T (7 kG), were made on specimens that had been stepwise demagnetized in AF's up to 100 mT. In general, the specimens having MDF's between 10 and 50 mT gave IRM curves that reached saturation after application of direct fields greater than 0.15 T (Figure 5, top). The moderate MDF's and low saturation field commonly indicate the presence of titanomagnetite, as would be expected in volcanic rocks. About a third of the specimens yielded MDF's greater than 50 mT and showed continual gains in IRM acquisition up to the 0.7-T limit of the experiment (Figure 5, bottom). In addition to titanomagnetite, these specimens contain a magnetic mineral having high coercivity, probably hematite or goethite.

Curie temperature determinations were made on small cores taken from the AF-demagnetized specimens described above. On heating up to 600°C the strong-field magnetization of these specimens gave curves as typified by Figures 6*a*–6*d*. In some cases (Figures 6*a* and 6*c*), the strong-field magnetization decreased sharply near 350°C, then increased to form a small hump between 400° and 500°C, and finally decreased to the paramagnetic base level between 500° and 585°C. On cooling, the strong-field magnetization increased sharply at about 500° to 580°C, finishing at room temperature at a much higher value than the initial strong-field magnetization. Because the experiment was run in an oxygen-free atmosphere, this type of curve indicates the presence of titanomaghemite, created by natural low-temperature oxidation of original titanomagnetite [*Marshall and Cox*, 1972]. The first sharp decrease in magnetization during heating gives the Curie temperature of the titanomaghemite. With continued heating, the increase in strong-field magnetization signals the inversion of titanomaghemite to titanium-poor titanomagnetite and a titanium-

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

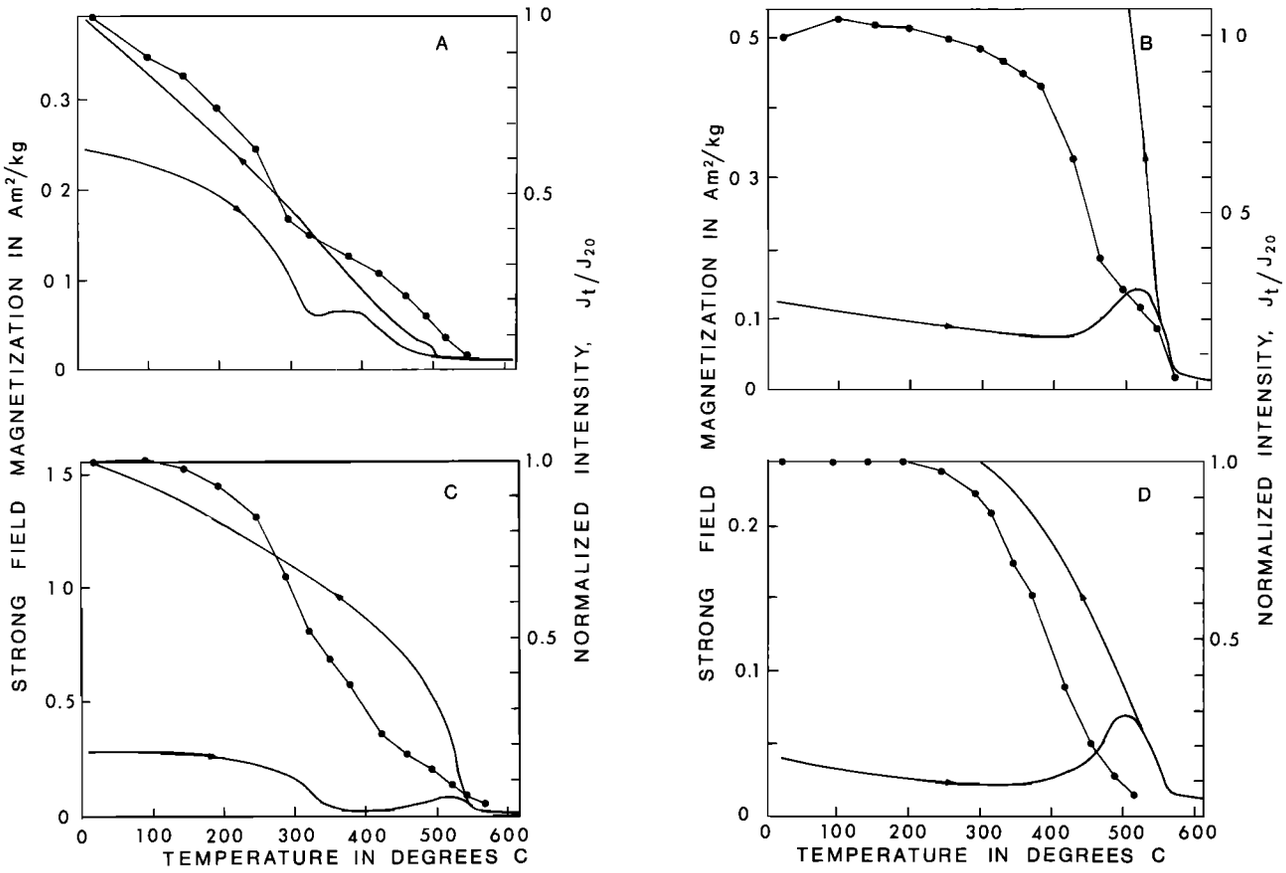


Fig. 6. Strong-field magnetization curves and thermal demagnetization (dotted curves) of NRM of representative specimens. (*A*) Site 4(3), Talkeetna Mountains; (*B*) Site 1(5), Talkeetna Mountains; (*C*) Site 10, Cantwell Formation; (*D*) Site 5, Cantwell Formation. For sites in Cantwell Formation, see Table 1 of *Hillhouse and Grommé*[1982].

rich phase such as ilmeno-hematite. The second sharp decrease in magnetization gives the Curie temperature of the newly created titanomagnetite phase. This new phase has a spontaneous magnetization that is much greater than the initial magnetization of the titanomaghemite. Figures 6*b* and 6*d* typify specimens in which the titanomaghemite Curie point is obscured by the onset of the inversion process.

We have plotted the decay of NRM as a function of temperature during thermal demagnetization for direct comparison with the strong-field magnetization curves. After heating in the furnace, these specimens, which were cut from the same cores that were used later in the Curie runs, were cooled in a near-zero field and then measured at room temperature. A sharp decrease in remanent magnetization typically occurred near the titanomaghemite Curie point (Figure 6*c*), but much of the natural remanence persisted until a temperature near the second Curie point had been reached. As outlined above, during the demagnetization experiment the titanomaghemite inverts to a cubic oxide phase (spinel structure) having a spontaneous magnetization that is greater than the magnetization of the starting material. As the inversion proceeds, the initial magnetic direction is carried and preserved by the new highly magnetic phase, as is shown by analysis of orthogonal projections of the magnetic components (Figure 7). We do not believe that the persistence of the remanent magnetization beyond the first Curie point is due to a second cubic phase that existed in the rock prior to the laboratory heating, because no evidence of such a phase appears in the strong-field magnetization curves (e.g., Figure 6*c*).

Magnetic components that were revealed by the thermal demagnetization experiments were analyzed according to the principal component method of *Kirschvink* [1980]. Free lines were fitted to the intermediate temperature steps corresponding to the titanomaghemite phase and to the high-temperature steps corresponding to the laboratory-induced cubic phase, as determined from the strong-field magnetization curves. Temperature steps below 200°C were excluded from this analysis to avoid contamination due to recently acquired viscous remanent magnetization. The angular difference between the intermediate- and high-temperature lines for a particular specimen rarely exceeded 10°, indicating that the laboratory-produced cubic phase inherited the original magnetic direction of the titanomaghemite. Because the angle between the intermediate temperature line and the line toward the origin from the intermediate line's centroid was always less than 10° and usually less than 5°, any secondary component possibly carried by hematite with a high unblocking temperature is negligible. For each specimen that was thermally demagnetized, a companion specimen (cut from the same core) was given progressive AF treatment up to 100 mT so that the effectiveness of the two cleaning techniques could be compared directly. Figures 7*a*–7*d* are typical examples from the 15 comparisons that were made for the Talkeetna and Teklanika volcanic rocks. We compared the intermediate temperature line to the optimum AF cleaning step as given in Table 2 of this study and in Table 1 of *Hillhouse and Grommé* [1982]. For each specimen the angular difference between the two magnetic directions rarely exceeded 10°, and systematic differences in di-

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

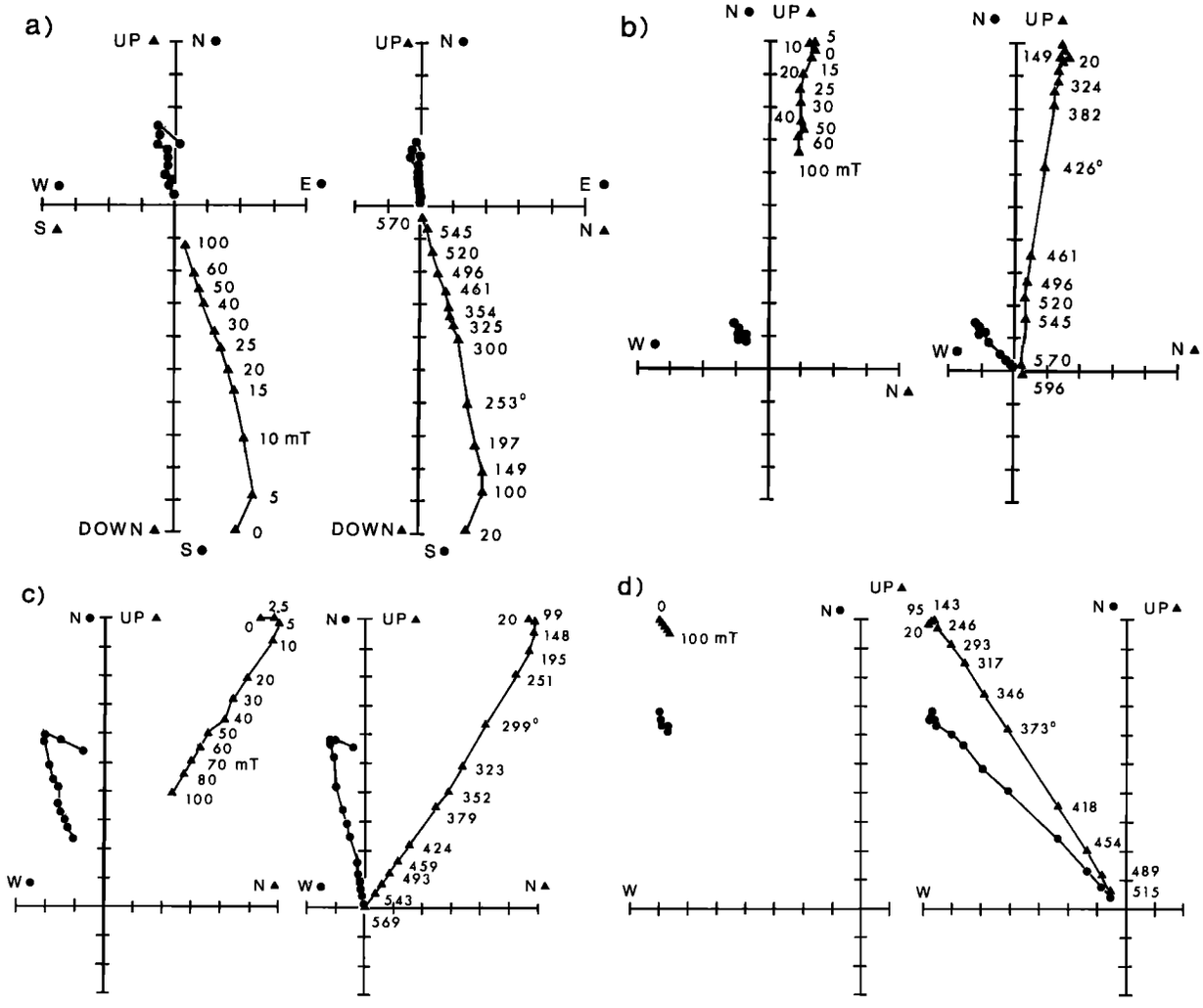


Fig. 7. Orthogonal projections of magnetic directions (not corrected for tilt of the bedding) following AF (left side) and thermal (right side) demagnetization of companion specimens to those in Figure 6. (*a*) AF: 1 division = 8.39×10^{-2} A/m, thermal: 1 division = 8.99×10^{-2} A/m; (*b*) AF: 1 division = 3.14×10^{-2} A/m, thermal: 1 division = 3.88×10^{-2} A/m; (*c*) AF: 1 division = 2.54×10^{-1} A/m, thermal: 1 division = 2.80×10^{-1}; (*d*) AF: 1 division = 9.80×10^{-3} A/m, thermal: 1 division = 8.26×10^{-3} A/m. Triangles denote vertical plane; dots denote horizontal plane.

rection were not apparent. Therefore we are confident that the AF treatment of both collections has effectively removed any secondary components of magnetization.

Examination of the polished sections showed that fresh titanomagnetite dominates the opaque mineralogy of the Talkeetna and Teklanika volcanic rocks. The titanomagnetite crystals commonly have large bladelike or skeletal shapes, although a few examples contained very fine equant grains. Evidence of low-temperature oxidation is indicated by whitened zones along cracks and around the margins of the titanomagnetite grains. We assume that titanomaghemite replaces the original material in the white zones. Reddish pigment, presumably hematite, clouds the area immediately surrounding some opaque grains, providing additional evidence of low-temperature oxidation. In thin section most of the andesite specimens contain a glass matrix and abundant plagioclase laths with hyalo-ophitic texture. Although primary igneous textures are preserved, most of the Teklanika specimens have cloudy plagioclase crystals, perhaps due to mild hydrothermal alteration.

From the polished sections, rock magnetic experiments, and the analysis of magnetic components we can propose a history of magnetization for these volcanic rocks. During original cooling, the lavas acquired thermoremanent magnetizations (TRM) carried by titanomagnetite crystals. The lavas were oxidized at temperatures low compared to the original Curie temperatures of the titanomagnetite, by weathering or perhaps also by mild hydrothermal activity. Low-temperature oxidation partially replaced ferrous ions with ferric ions, leaving vacancies in the octahedral lattice of the titanomagnetite and thereby partially transforming the original magnetic crystals to titanomaghemite. Oxidation experiments on submarine basalts show that the alteration to titanomaghemite preserves the original TRM direction when the oxidation occurs at temperatures well below the original Curie point [*Marshall and Cox*, 1971]. Whether this result holds for subaerial basalts under natural conditions remains to be tested. The rocks acquired minor amounts of high-coercivity magnetic minerals, such as the weathering product, goethite, or the oxidation product, hematite, as indicated by the IRM experiments. After AF treatment, any contribution that these high-coercivity magnetic constituents make to the total magnetization are negligible, as demonstrated by the principal component analysis.

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TECTONIC IMPLICATIONS OF THE PALEOMAGNETIC DATA

In this section we compare paleomagnetic poles from south central Alaska with reference poles from North America to test for Tertiary displacement of the Talkeetna superterrane (Figure 8). Table 3 summarizes the pertinent paleomagnetic data for Alaska and the undisturbed interior of North America. The Alaskan data include paleomagnetic poles from the Paleocene Cantwell Formation [*Hillhouse and Grommé*, 1982], the volcanic rocks of the Paleocene Nowitna unit near McGrath [*Plumley*, 1984], and Paleocene or Eocene sandstones of the Chickaloon Formation in Matanuska Valley [*Stone*, 1984]. Early Tertiary reference poles were derived from igneous rocks located primarily in Montana and Wyoming [*Diehl et al.*, 1983]. In calculating the paleolatitude of each study area except the Nowitna unit we have used the mean of individual site VGP's, with 95% confidence limits on the paleolatitudes given according to the method of *Demarest* [1983]. The two right-hand columns of the table are measures of rotation and latitude displacement of the study areas relative to the appropriate reference pole. Ninety-five-percent confidence limits, which combine the uncertainties in the Alaskan poles and the reference poles, were calculated by the method of *Wilson and Cox* [1980]. The displacement and rotation listed for the Nowitna unit were obtained from *Coe et al.* [1985].

The paleolatitude (76°N) of the northern Talkeetna Mountains is 8° ± 10° higher than would be predicted had the site remained fixed relative to North America since the early Eocene, implying that the region has drifted south relative to the craton. Therefore the large-scale northward drift of the Talkeetna superterrane, as indicated by significantly low paleolatitudes of its Jurassic and Triassic rocks [*Hillhouse and Grommé*, 1984; *Stone et al.*, 1982], must have been completed before 50 m.y. ago. For simplicity, we refer to relative increases in paleolatitude as "northward" drift, keeping in mind that the drift is relative to a fixed North America, not the present geographic coordinate system.

The northern Talkeetna paleolatitude anomaly is in accord with the paleolatitudes that were determined from the Nowitna unit rocks and the Cantwell Formation, both located north of the Denali fault (Figure 1). All results suggest modest southward drift of the region relative to the craton. We can extend these kinematic constraints beyond the study areas to the rest of the overlapped basement terranes, in particular, the Wrangellia, Peninsular, Yukon-Tanana, and Nixon Fork terranes. In terms of large-scale movements the entire region encompassed by these terranes has apparently acted as a coherent block since the early Tertiary. The region is bounded on the north by the Tintina and Kaltag faults and extends southward as far as the Border Ranges fault, the northern limit of the Chugach terrane. Although numerous faults having late Tertiary displacements lie within this region, geologic mapping indicates that the displacements are no greater than a few hundred kilometers. The magnitude of Tertiary lateral movement along the Denali fault is somewhat controversial [*Csejtey et al.*, 1982; *Lanphere*, 1978; *Nokleberg*, 1983], with estimates ranging from just a few kilometers up to 500 km.

Paleomagnetic results from the Paleocene or Eocene Chickaloon Formation [*Stone*, 1984], a continental sandstone sequence in the Matanuska Valley (Figure 1; Table 3), yielded a paleolatitude anomaly of 46° ± 17°, which does not agree

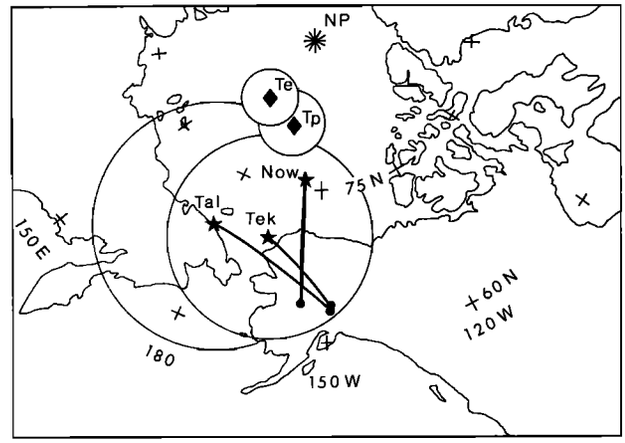


Fig. 8. Mean paleomagnetic poles from lower Tertiary volcanic rocks of Alaska are compared with North American reference poles. Eocene (*Te*) and Paleocene (*Tp*) reference poles are shown with their 95% confidence limits (see Table 3). (*Tal*) Talkeetna Mountains volcanic rocks (Early Eocene); (*Tek*) Volcanic rocks of the upper part (= Teklanika Formation of *Gilbert et al.* [1976] of the Cantwell Formation (Paleocene); (*Now*) Volcanic rocks of the Paleocene Nowitna unit [*Plumley*, 1984]; (*NP*) present geographic pole.

with the results from the more northerly volcanic sites. The Chickaloon overlies the Upper Jurassic and Cretaceous overlap assemblage at the southern edge of the Peninsular terrane, the same assemblage that underlies lower Tertiary volcanic rocks of the Talkeetna Mountains. Therefore, on geologic grounds the Chickaloon and Talkeetna volcanic rocks should have a common translational history. Explaining the difference in the paleomagnetic results by proposing thousands of kilometers of displacement on the Castle Mountain fault, which bounds the Chickaloon Formation on the north, is contradicted by mapped offsets of a few tens of kilometers in the Upper Cretaceous sedimentary rocks [*Detterman et al.*, 1974; *Grantz*, 1966]. Either the Chickaloon paleolatitude is invalid, perhaps owing to systematic flattening of the inclination in the sediments, or an unknown Tertiary suture exists in the vicinity of the Castle Mountain fault.

In southwest Alaska, paleomagnetic poles from uppermost Cretaceous (68 m.y.) volcanic rocks yield a paleolatitude of 65° ± 4° and a latitude anomaly of 9° ± 7° [*Globerman and Coe*, 1983; *Coe et al.*, 1985]. The samples were collected from the vicinity of Hagemeister Island in northern Bristol Bay. *Thrupp and Coe* [1984] reported no significant latitude anomaly from Paleocene and Eocene volcanic rocks in the vicinity of Lake Clark and Iliamna Lake. Paleolatitude anomalies in southwestern Alaska and in the northern Talkeetna Mountains are hardly significant, in view of the confidence limits. In contrast, paleolatitude anomalies reported from Paleocene and Eocene rocks on the southwestern tip of the Alaska Peninsula are much greater, implying an average northward displacement of 2500 km [*Stone et al.*, 1982]. As with the Chickaloon, these results seem to imply that a suture exists in the southern margin of the Peninsular terrane, perhaps along the Bruin Bay fault. The thin arcuate region between the Bruin Bay and Border Ranges faults presents an enigma because its basement is tied geologically to the northern terranes and yet its paleolatitudes are evidently too low for it to be a part of the northern terranes in the early Tertiary. Paleolatitude determinations from this arcuate region are highly inconsistent with one another and may be contaminated by secondary magnetizations, but further rock magnetic investigations are needed to resolve this question.

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3. Mean Virtual Geomagnetic Poles of Early Tertiary Age From Alaska and Central North America

No.	Study	Age	N	Mean VGP Latitude, °N	Mean VGP Longitude, °E	α_{95}, deg	Paleolatitude, °N	R, deg	D, deg	Reference
1	Northern Talkeetna Mountains, volcanic rocks (63.1°N, 211.0°E)	Early Eocene	26	69.5	179.6	12.2	76 ± 10	−37 ± 44	−8 ± 10	This study
2	Volcanic rocks of the upper Cantwell Formation 63.6°N, 211.0°E)	Paleocene	18	70.0	194.6	10.0	81 ± 8	−29 ± ??	−9 ± 8	*Hillhouse and Grommé* [1982]
3	Volcanic rocks, Nowitna unit (63.8°N, 204.7°E)	Paleocene	6	76.9	198.1	25.3	75 ± 20	5 ± ??	−1 ± 22	*Plumley* [1984]
4	Chickaloon Formation (61.8°N, 211.5°E)	Eocene or Paleocene	4	51.5	32.2	21.8	23 ± 17	7 ± 20	46 ± 17	*Stone* [1984]
				North America Reference Poles						
5	Middle and early Eocene	45–54 m.y.	147	82.8	170.4	3.0				*Diehl et al.* [1983]
6	Paleocene	55–67 m.y.	68	81.5	192.6	3.2				*Diehl et al.* [1983]
7	Early Tertiary (Poles 5 & 6 combined)	45–67 m.y.	215	82.5	178.4	2.3				

Notes: α_{95}, 95% confidence limit for mean pole; R, paleodeclination anomaly, positive clockwise; D, paleolatitude anomaly, negative value implies southward displacement; N, number of units averaged; ??, indeterminate because confidence limit encloses site.

Another body of paleomagnetic data that tends to support large-scale displacement of the southern rim of the Peninsular terrane comes from the Chugach and Prince William terranes to the south. Upper Cretaceous volcanic rocks on Resurrection Peninsula near Seward [*Grommé and Hillhouse*, 1981] and Paleocene volcanic rocks from Kodiak Island [*Plumley et al.*, 1983] yield paleolatitude anomalies suggesting approximately 30° of northward drift relative to the craton. Paleolatitudes from sedimentary rocks of the southern terranes are also significantly low, although the agreement between sites is poor [*Stone et al.*, 1982]. *Moore et al.* [1983] have proposed several models involving substantial right-lateral slip on faults in southern Alaska to account for the paleolatitude anomalies of the southern terranes. The most reliable paleomagnetic evidence, which is from the Tertiary volcanic rocks, shows that the models are only applicable to the region south of the Bruin Bay and Castle Mountain faults.

The Talkeetna and Teklanika poles lie south of the Tertiary reference poles (Figure 8), implying that the region has rotated about 30° counterclockwise relative to the craton. However, the confidence limits on the apparent rotation are too large to be useful, because our Alaskan sites have paleocolatitudes that are nearly equal to the 95% confidence limits of the poles (Table 3). Nevertheless, our results are consistent with other Late Cretaceous and Paleocene poles from volcanic rocks in southwestern Alaska, which show a statistically significant counterclockwise rotation (Figure 9). *Globerman and Coe* [1983] first proposed such a rotation on the basis of paleomagnetic results from the volcanic section in Northern Bristol Bay. Dated at 68 m.y., the flows yielded a counterclockwise paleodeclination anomaly of 43° ± 23°. G. A. Thrupp obtained a counterclockwise anomaly of 54° ± 22° from approximately 66-m.y.-old lava flows near Lake Clark [*Coe et al.*, 1985]. Younger flows (44 m.y.) in the Lake Clark region gave a counterclockwise paleodeclination anomaly of 12° ± 42°. Taken together, Upper Cretaceous and lower Tertiary volcanic rocks distributed from Bristol Bay to the Talkeetna mountains give paleodeclinations consistent with a 40° counterclockwise rotation of western Alaska. In contrast, sites south of the Bruin Bay and Castle Mountain faults yield paleodeclinations that do not follow this simple pattern.

A tectonic model which leaves south central Alaska fixed in latitude but free to rotate counterclockwise relative to North America after 50 m.y. ago could explain the paleomagnetic results from the region north of the Bruin Bay fault. Substantial right-lateral motion on the Tintina fault, as proposed by *Gabrielse* [1985], would impart the proper magnitude and sense of rotation but would imply a small northward displacement of the region. We tested this idea by fitting the Tintina-Kaltag fault system to a small circle (radius = 22°) and then computed the pole (43°N, 150°W) and the amount of rotation (24°) needed to produce a 1000-km right-lateral offset on the fault system. Application of this rotation to the mean Alaskan site predicts a 37° counterclockwise paleodeclination anomaly and 5° of northward drift, using reference pole 7 in Table 3. This paleodeclination anomaly fits the Talkeetna data well, but the latitude anomaly predicts more northward drift than is observed in the Talkeetna paleomagnetic data. A major question concerning this model is the connection between the Kaltag and Tintina faults. Much of the motion along the Tintina may have been taken up by compression in central Alaska rather than by motion on the Kaltag fault. Another problem with this model is the timing of motion along the Tintina fault, where much of the displacement occurred in pre-Tertiary time.

An alternative model was proposed by *Globerman and Coe* [1983, 1984], who noted that the early Tertiary poles are consistent with models for oroclinal bending of Alaska [*Carey*,

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1955; *Freeland and Dietz*, 1973]. In general, these models call for southward drift of southwestern Alaska as the region underwent counterclockwise rotation about a hinge line near the 145°W meridian. An objection to this hypothesis is raised by the nearness of the hinge line to the Teklanika and Cantwell sites, which would predict greater southward translations for sites located farther west of the hinge line. The paleomagnetic data indicate the opposite trend in paleolatitude anomalies, although the differences are barely significant at the 95% confidence level. Also, the original oroclinal models postulate substantial seafloor spreading in the Canada Basin of the Arctic Ocean during Tertiary time in order to accommodate the rotation. However, geophysical evidence suggests that the oceanic crust in the basin is much older than 50 m.y. [*Sweeney*, 1983; *Lawver and Baggeroer*, 1983], and was formed before the rotation of southwestern Alaska occurred. *Coe et al.* [1985] favored a variant of the oroclinal bending hypothesis following *Grantz* [1966] and *Patton and Tailleur* [1977], in which northern Alaska was decoupled from southwestern Alaska during oroclinal bending. The driving force for the rotation might have been seafloor spreading in the North Atlantic and Eurasian basin of the Arctic.

Additional studies of the early Tertiary of northern Alaska and the northeast USSR are needed to determine whether the tectonic implications of the central Alaska poles are applicable to the broader Amerasian Arctic region. If so, this region may comprise a small plate caught between North America and Eurasia, and its motion would have been influenced by the relative motions of the continental plates. The small plate would include southwestern Alaska and northeasternmost USSR. Beginning in the early Tertiary, seafloor spreading along the Gakkel and northern Atlantic ridges might thus have forced changes in the small plate's position relative to Eurasia and North America.

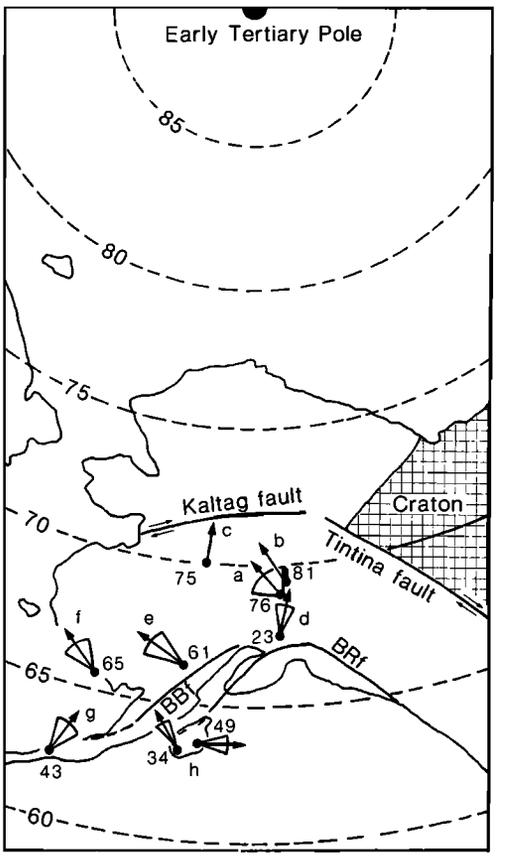


Fig. 9. Paleolatitude (numbers) and paleodeclination (arrows with 95% confidence limits) determinations from lower Tertiary and Upper Cretaceous rocks in Alaska are compared to the early Tertiary paleolatitude grid for nuclear North America. The grid is projected about the early Tertiary reference pole given in Table 3. Paleomagnetic studies: (*a*) Northern Talkeetna Mountains (this study), (*b*) Cantwell Formation [*Hillhouse and Grommé*, 1982], (*c*) Volcanic rocks of the Nowitna unit [*Plumley*, 1984], (*d*) Chickaloon Formation (Eocene or Paleocene) [*Stone*, 1984], (*e*) Lake Clark–Lake Iliamna volcanic rocks (Paleocene) [*Coe et al.*, 1985], (*f*) Northern Bristol Bay volcanic series (Upper Cretaceous) [*Coe et al.*, 1985], (*g*) Tolstoi Formation (Paleocene and Eocene) [*Stone et al.*, 1982], (*g*) Ghost Rocks Formation (Eocene and Paleocene) [*Plumley et al.*, 1983]. Confidence limits on the paleodeclinations for studies *b* and *c* are indeterminate because the confidence circles enclose the sites. (*BBf*) Bruin Bay fault, (*BRf*) Border Ranges fault.

CONCLUSIONS

In the northern Talkeetna Mountains, early Eocene (and possibly Paleocene) volcanic rocks overlap the boundaries of major Alaskan accretionary terranes, namely the Wrangellia and Peninsular terranes. Paleomagnetic studies were made along a 75-km transect from the Denali fault to the Talkeetna River. Andesite flows and a dacite body were sampled at eight localities for a total of 26 cooling units. With the exception of one site, AF treatments were sufficient to isolate well-defined stable end point magnetizations. Curie temperature determinations and analysis of IRM acquisition curves indicate that the original magnetic mineral was titanomagnetite, commonly produced during the initial cooling of subaerial lavas. Most of the lavas have undergone low-temperature oxidation to produce titanomaghemite from original titanomagnetite. Curie temperatures of the titanomaghemite range from 325° to 550°C. Magnetic directions are mainly of reversed polarity, although three cooling units are normally polarized. Stratigraphic sections from the opposite limbs of a syncline yield directional paths that converge when corrections are made for tilt of the bedding, satisfying a necessary condition for primary origin of the magnetization. The mean of all unit virtual geomagnetic poles is a pole at 69.5°N, 179.6°E, with an $\alpha_{95} = 12.2°$.

The corresponding mean paleolatitude (76°N) of the northern Talkeetna Mountains is 8° ± 10° higher than the latitude predicted from the North American reference pole [*Diehl et al.*, 1983]. Therefore, northward drift of the Talkeetna superterrane, as indicated by paleomagnetic studies of the Mesozoic rocks of Wrangellia [*Hillhouse and Grommé*, 1984] and the Peninsular terrane [*Stone et al.*, 1982], was essentially complete by 50 m.y. ago. This conclusion does not agree with the significantly low paleolatitudes that Stone and his coworkers reported from Paleocene sedimentary rocks in the southwestern and southeastern parts of the Peninsular terrane. The narrow, arcuate region between the Bruin Bay–Castle Mountain fault system and the Border Ranges fault presents an enigma, because the Mesozoic basement rocks are geologically tied to the northern terranes, while the region's early Tertiary paleolatitudes seem to be too low to be compatible with results from the northern Talkeetna volcanic rocks. Paleolatitude determinations from the sedimentary rocks show very poor internal consistency, and may be contaminated by secondary components of magnetization. If further rock magnetic experiments support the validity of the low paleolatitudes, then a Tertiary suture must be placed near the Bruin Bay–Castle Mountain fault system to close a latitude gap of ap-

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

proximately 30°. Terranes south of the Border Ranges fault, such as the Chugach and the Prince William, also give low paleolatitudes implying post-Paleocene northward drift of approximately 30°.

With the exception of the enigmatic southern rim of the Peninsular terrane early Tertiary paleomagnetic poles from the Talkeetna superterrane, the Yukon-Tanana terrane, and the Nixon Fork terrane indicate that the region was a coherent block since the early Tertiary. The northern boundary of this region is currently drawn at the Kaltag and Tintina faults, simply because paleomagnetic studies of the early Tertiary have not been carried out north of this line. Numerous faults having late Tertiary movement occur within the region, although the displacements are probably no greater than a few hundred kilometers. Paleomagnetic poles from volcanic rocks in the northern Talkeetna Mountains, Denali National Park, Lake Clark, Bristol Bay, and near McGrath generally lie south of the reference poles, indicating that the region has rotated counterclockwise about 40° relative to North America since the Paleocene.

From the paleomagnetic evidence we infer that early Tertiary (about 65–50 m.y.) volcanism followed the final consolidation of the far-traveled terranes to form a coherent tectonic block in south central Alaska. Wrangellia's northward drift essentially had been completed by the end of the Paleocene. The consolidated block was probably in place when the southernmost terranes (Chugach and Prince William) accreted to the continental margin in post-Paleocene time.

Acknowledgments. We thank Victoria Pease for performing the magnetic measurements. We also thank J. L. Kirschvink for lending us his only machine-readable copy of his principal component analysis programs, and J. T. Hagstrum for adaptation of these programs. Clyde Wahrhaftig, Rob Coe, Gordon Thrupp, and Brian Globerman offered many helpful suggestions for improving the manuscript.

REFERENCES

Carey, S. W., The orocline concept in geotectonics, *R. Soc. Tasmania Pap. Proc.*, *89*, 255–288, 1955.

Coe, R. S., B. R. Globerman, P. W. Plumley, and G. A. Thrupp, Paleomagnetic results from Alaska and their tectonic implications, in *Tectonostratigraphic Terranes of the Circum-Pacific Region*, *AAPG Circum-Pac. Earth Sci. Ser.*, 1, edited by D. G. Howell, pp. 85–108, American Association of Petroleum Geologists, Tulsa, Okla., 1985.

Cox, A., Confidence limits for the precision parameter κ, *Geophys. J. R. Astron. Soc.*, *18*, 545–549, 1969.

Cox, A., Latitude dependence of the angular dispersion of the geomagnetic field, *Geophys. J. R. Astron. Soc.*, *20*, 253–269, 1970.

Cox, A., and R. G. Gordon, Paleolatitudes determined from paleomagnetic data from vertical cores, *Rev. Geophys.*, *22*, 47–72, 1984.

Csejtey, B., Jr., W. H. Nelson, D. L. Jones, N. J. Silberling, R. M. Dean, M. S. Morris, M. A. Lanphere, J. G. Smith, and M. L. Silberman, Reconnaissance geologic map and geochronology, Talkeetna Mountains quadrangle, northern part of Anchorage quadrangle, and southwest corner of Healy quadrangle, Alaska, scale 1:250,000, *U.S. Geol. Surv. Open file Rep.*, *78-558-A*, 1–62, 1978.

Csejtey, B., Jr., H. L. Foster, and W. J. Nokleberg, Cretaceous accretion of the Talkeetna superterrane and subsequent development of the Denali fault in south-central and eastern Alaska (abstract), *Geol. Soc Am. Abstr. Programs*, *12(7)*, 409, 1980.

Csejtey, B., Jr., D. P. Cox, R. C. Evarts, G. D. Stricker, and H. L. Foster, The Cenozoic Denali fault system and the Cretaceous accretionary development of southern Alaska, *J. Geophys. Res.*, *87*, 3741–3754, 1982.

Demarest, H. H., Jr., Error analysis for the determination of tectonic rotation from paleomagnetic data, *J. Geophys. Res.*, *88*, 4321–4328, 1983.

Detterman, R. L., G. Plafker, T. Hudson, R. G. Tysdal, and N. Pavoni, Surface geology and Holocene breaks along the Susitna segment of the Castle Mountain fault, Alaska (scale 1:24,000), *U.S. Geol. Surv. Field Stud. Map*, *MF-618*, 1974.

Diehl, J. F., M. E. Beck, Jr., S. Beske-Diehl, D. Jacobson, and B. C. Hearn, Jr., Paleomagnetism of the Late Cretaceous-early Tertiary north-central Montana alkalic province, *J. Geophys. Res.*, *88*, 10593–10609, 1983.

Freeland, G. L., and R. S. Dietz, Rotation history of Alaskan tectonic blocks, *Tectonophysics*, *18*, 379–389, 1973.

Gabrielse, H., Major dextral transcurrent displacements along the northern Rocky Mountain trench and related lineaments in north-central British Columbia, *Geol. Soc. Am. Bull.*, *96*, 1–14, 1985.

Gilbert, W. G., Evidence for early Cenozoic orogeny in central Alaska Range, *Geol. Rep. Alaska Div. Geol. Geophys. Surv.*, *51*, 5–7, 1976.

Gilbert, W. G., V. M. Ferrell, and D. L. Turner, The Teklanika Formation—A new Paleocene volcanic formation in the central Alaska Range, *Geol. Rep. Alaska Div. Geol. Geophys. Surv.*, *47*, 1–16, 1976.

Globerman, B. R., and R. S. Coe, Tectonic implications of paleomagnetic data from northern Bristol Bay, SW Alaska (abstract), *Eos Trans. AGU*, *64*, 688, 1983.

Globerman, B. R., and R. S. Coe, Discordant declinations from "in place" Upper Cretaceous volcanics, SW Alaska: Evidence for oroclinal bending? (abstract), *Geol. Soc. Am. Abstract Progr.*, *16(5)*, 286, 1984.

Grantz, A., Strike-slip faults in Alaska, *U.S. Geol. Surv. Open File Rep.*, 1–82, 1966.

Grommé, S., and J. W. Hillhouse, Paleomagnetic evidence for northward movement of the Chugach terrane, southern and southeastern Alaska, *U.S. Geol. Surv. Circ.*, *823-B*, B70–B72, 1981.

Harrison, C. G. A., Secular variation and excursions of the earth's magnetic field, *J. Geophys. Res.*, *85*, 3511–3522, 1980.

Helsley, C. E., Paleomagnetic evidence for the rotation of Alaska, *N.T.I.S. Open File Rep.*, *N70-42234*, 1–21, 1970.

Hickman, R. G., and C. Craddock, Geologic map of central Healy quadrangle, Alaska, scale 1:63,360, *Alaska Open File Rep.*, *95*, 1976.

Hillhouse, J. W., Paleomagnetism of the Triassic Nikolai Greenstone, McCarthy quadrangle, Alaska, *Can. J. Earth Sci.*, *14*, 2578–2592, 1977.

Hillhouse, J. W., and C. S. Grommé, Limits to northward drift of the Paleocene Cantwell Formation, central Alaska, *Geology*, *10*, 552–556, 1982.

Hillhouse, J. W., and C. S. Grommé, Wrangellia in southern Alaska 50 m.y. ago, *Eos Trans. AGU*, *64*, 687, 1983.

Hillhouse, J. W., and C. S. Grommé, Northward displacement and accretion of Wrangellia: New paleomagnetic evidence from Alaska, *J. Geophys. Res.*, *89*, 4461–4477, 1984.

Jones, D. L., N. J. Silberling, and J. Hillhouse, Wrangellia—A displaced terrane in northwestern North America, *Can. J. Earth Sci.*, *14*, 2565–2577, 1977.

Jones, D. L., N. J. Silberling, Bela Csejtey, Jr., W. H. Nelson, and C. D. Blome, Age and structural significance of ophiolite and adjoining rocks in the Upper Chulitna District, south-central Alaska, *U.S. Geol. Surv. Prof. Pap.*, *1121-A*, A1–A21, 1980.

Jones, D. L., N. J. Silberling, H. C. Berg, and G. Plafker, Map showing tectonostratigraphic terranes of Alaska, columnar sections, and summary description of terranes, scale 1:2,500,000, *U.S. Geol. Surv. Open File Rep.*, *81-792*, 1–20, 1981.

Jones, D. L., N. J. Silberling, W. Gilbert, and P. Coney, Character, distribution, and tectonic significance of accretionary terranes in the central Alaska Range, *J. Geophys. Res.*, *87*, 3709–3717, 1982.

Kirschvink, J. L., The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699–718, 1980.

Lanphere, M. A., Displacement history of the Denali fault system, Alaska and Canada, *Can. J. Earth Sci.*, *15*, 817–822, 1978.

Lawver, L. A., and A. Baggeroer, A note on the age of the Canada Basin, *J. Alaska Geol. Soc.*, *2*, 57–66, 1983.

Marshall, M., and A. Cox, Effect of oxidation on the natural remanent magnetization of titanomagnetite in suboceanic basalt, *Nature*, *230*, 28–31, 1971.

Marshall, M., and A. Cox, Magnetic changes in pillow basalt due to sea floor weathering, *J. Geophys. Res.*, *77*, 6459–6469, 1972.

McElhinny, M. W., and R. T. Merrill, Geomagnetic secular variation over the past 5 m.y., *Rev. Geophys.*, *13*, 687–708, 1975.

Moore, J. C., T. Byrne, P. W. Plumley, M. Reid, H. Gibbons, and R. S. Coe, The Paleocene evolution of the Kodiak Islands, Alaska: Consequences of ridge-trench interaction in a more southerly latitude, *Tectonics*, *2*, 265–293, 1983.

Nokleberg, W. J., Cretaceous accretion of Wrangellia along the Broxson Gulch thrust and Cenozoic strike-slip displacement of terranes along the Denali fault in southern Alaska (abstract), paper present-

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

ed at 58th Annual Meeting of American Association of Petroleum Geologists, Society of Exploration Geophysicists, Society of Economic Paleontologists and Mineralogists, Pacific sections, Sacramento, Calif., 1983.

Packer, D. R., and D. B. Stone, Paleomagnetism of Jurassic rocks from southern Alaska, and the tectonic implications, *Can. J. Earth Sci.*, *11*, 976–997, 1974.

Patton, W. W., and I. L. Tailleur, Evidence in the Bering Strait region for differential movement between North America and Eurasia, *Geol. Soc. Am. Bull.*, *88*, 1298–1304, 1977.

Plumley, P. W., A paleomagnetic study of the Prince William terrane and Nixon Fork terrane, Alaska, Ph.D. thesis, 190 pp., Univ. of California, Santa Cruz, 1984.

Plumley, P. W., and R. S. Coe, Paleomagnetic investigation of the Nixon Fork terrane lower Paleozoic section, paper presented at 58th Annual Meeting of American Association of Petroleum Geologists, Society of Exploration Geophysicists, Society of Economic Paleontologists and Mineralogists Pacific sections, Sacramento, Calif., 1983.

Plumley, P. W., R. S. Coe, and Tim Byrne, Paleomagnetism of the Paleocene Ghost Rocks Formation, Prince William terrane, Alaska, *Tectonics*, *2*, 295–314, 1983.

Reed, B. L., and M. A. Lanphere, Generalized geologic map of the Alaska–Aleutian Range batholith showing potassium-argon ages of the plutonic rocks, *U.S. Geol. Surv. Misc. Field Stud. Map*, *MF-372*, 1972.

Reed, B. L., and S. W. Nelson, Geologic map of the Talkeetna quadrangle, Alaska, scale 1:250,000, *U.S. Geol. Surv. Misc. Invest. Ser.*, *I-1174*, 1980.

Stone, D. B., Triassic paleomagnetic data and paleolatitudes for Wrangellia, Alaska, *Geol. Rep. Alaska Div. Geol. Geophys. Surv.*, *73*, 55–62, 1981.

Stone, D. B., Concordance and conflicts in Alaskan paleomagnetic data, *Stanford Univ. Publ. Geol. Sci.*, *18*, 186–189, 1984.

Stone, D. B., and D. R. Packer, Tectonic implications of Alaska Peninsula paleomagnetic data, *Tectonophysics*, *37*, 183–201, 1977.

Stone, D. B., and D. R. Packer, Paleomagnetic data from the Alaska Peninsula, *Geol. Soc. Am. Bull.*, *90*, 545–560, 1979.

Stone, D. B., B. C. Panuska, and D. R. Packer, Paleolatitudes versus time for southern Alaska, *J. Geophys. Res.*, *87*, 3697–3707, 1982.

Sweeney, R. F., Evidence for the origin of the Canada Basin margin by rifting in Early Cretaceous time, *J. Alaska Geol. Soc.*, *2*, 17–23, 1983.

Thrupp, G. A., and R. S. Coe, Paleomagnetism of Tertiary volcanic rocks from the Alaska Peninsula: Is the Peninsular terrane a coherent tectonic entity?, *Geol. Soc. Am. Abstr. Programs*, *16*, 337, 1984.

Wahrhaftig, C., Quaternary geology of the Nenana River valley and adjacent parts of the Alaska Range, *U.S. Geol. Surv. Prof. Paper*, *293A*, 1–68, 1958.

Wahrhaftig, C., J. A. Wolfe, E. B. Leopold, and M. A. Lanphere, The coal-bearing group in the Nenana coal field, Alaska, *U.S. Geol. Surv. Bull.*, *1274-D*, D1–D30, 1969.

Wilson, D., and A. Cox, Paleomagnetic evidence for tectonic rotation of Jurassic plutons in Blue Mountains, eastern Oregon, *J. Geophys. Res.*, *85*, 3681–3689, 1980.

Wolfe, J. A., and C. Wahrhaftig, The Cantwell Formation of the central Alaska Range, *U.S. Geol. Surv. Bull.*, *1294-A*, A41–A46, 1970.

B. Csejtey, Jr., C. S. Grommé, and J. W. Hillhouse, U.S. Geological Survey, Menlo Park, CA 94025.

(Received March 12, 1985;
revised July 26, 1985;
accepted July 26, 1985.)

21562202b, 1985, B14, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB090iB14p12523 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License